

Ultrasonic Sensor Accuracy & Noise Characterization

Objective

The goal of this investigation was to characterize the accuracy, noise, and short-term stability of an HC-SR04 ultrasonic distance sensor. the kind of bench characterization work used to understand how a sensor behaves before it's trusted in a larger system. This mirrors, at a small scale, the process of evaluating a sensor's noise, drift, and stability against known reference values.

Method

An HC-SR04 ultrasonic sensor was connected to an Arduino Uno R3 and triggered continuously to measure distance. Each reading was timestamped and streamed over serial to a computer, where a Python script logged the data to CSV files and computed summary statistics.

Two types of tests were run:

1. **Accuracy sweep:** a flat target was placed at six known true distances (5, 10, 15, 20, 25, and 30 cm, measured with a ruler), and roughly 10 seconds of readings were logged at each distance.
2. **Stability test:** the target was left untouched at a fixed distance (20 cm) for an extended period (5 minutes, 2,997 samples) to check whether readings drifted over time even with nothing physically changing.

For each dataset, the mean distance, error relative to true distance, and noise (standard deviation) were computed. For the stability test, a drift rate (cm/min) was also computed by fitting a line to distance over time.

Results

Accuracy sweep results:

True Distance (cm)	Measured Mean (cm)	Error	Noise STD dev
5	5.17	0.17	0.010
10	10.37	0.37	0.010
15	14.61	-0.39	0.006
20	19.07	-0.93	0.296
25	24.03	-0.97	0.292
30	29.00	-1.00	0.339

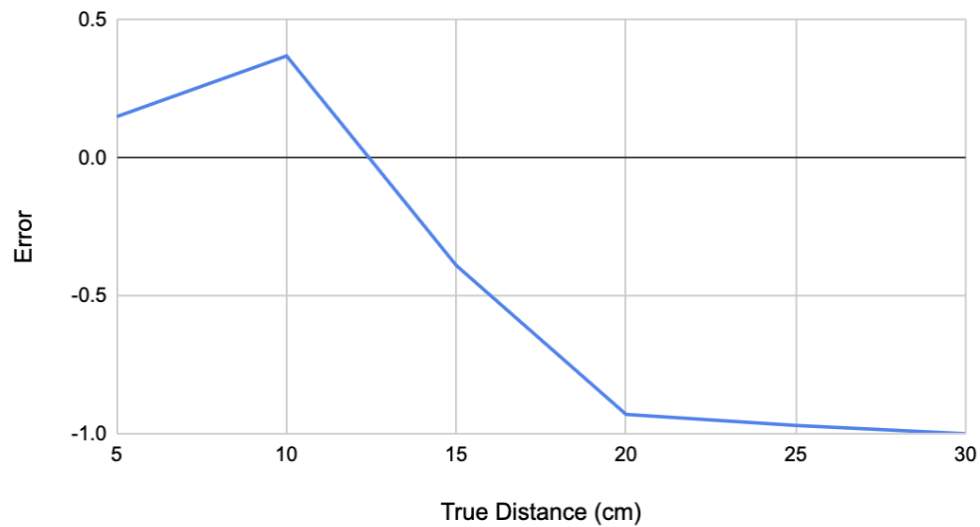
(Error = Measured Mean – True Distance)

Stability test results (target fixed at 20 cm, 5 minute duration):

Metrics	Value
Samples	2997
Mean Distances	19.17 cm
Noise STD	0.006 cm
Min / Max	19.16 cm / 19.19 cm
Drift Rate	-0.0002 cm/min

Error versus True Distance Chart

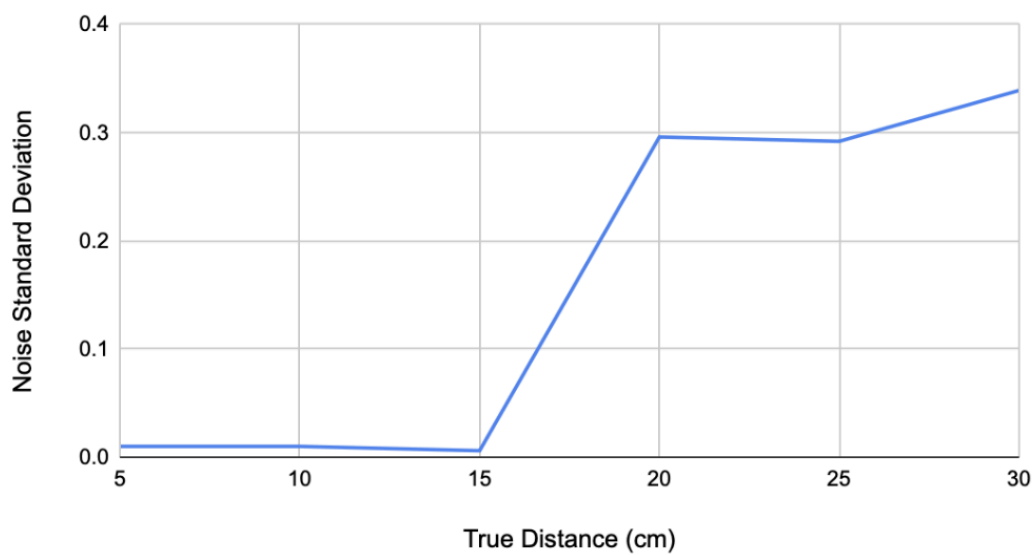
Error vs. True Distance (cm)



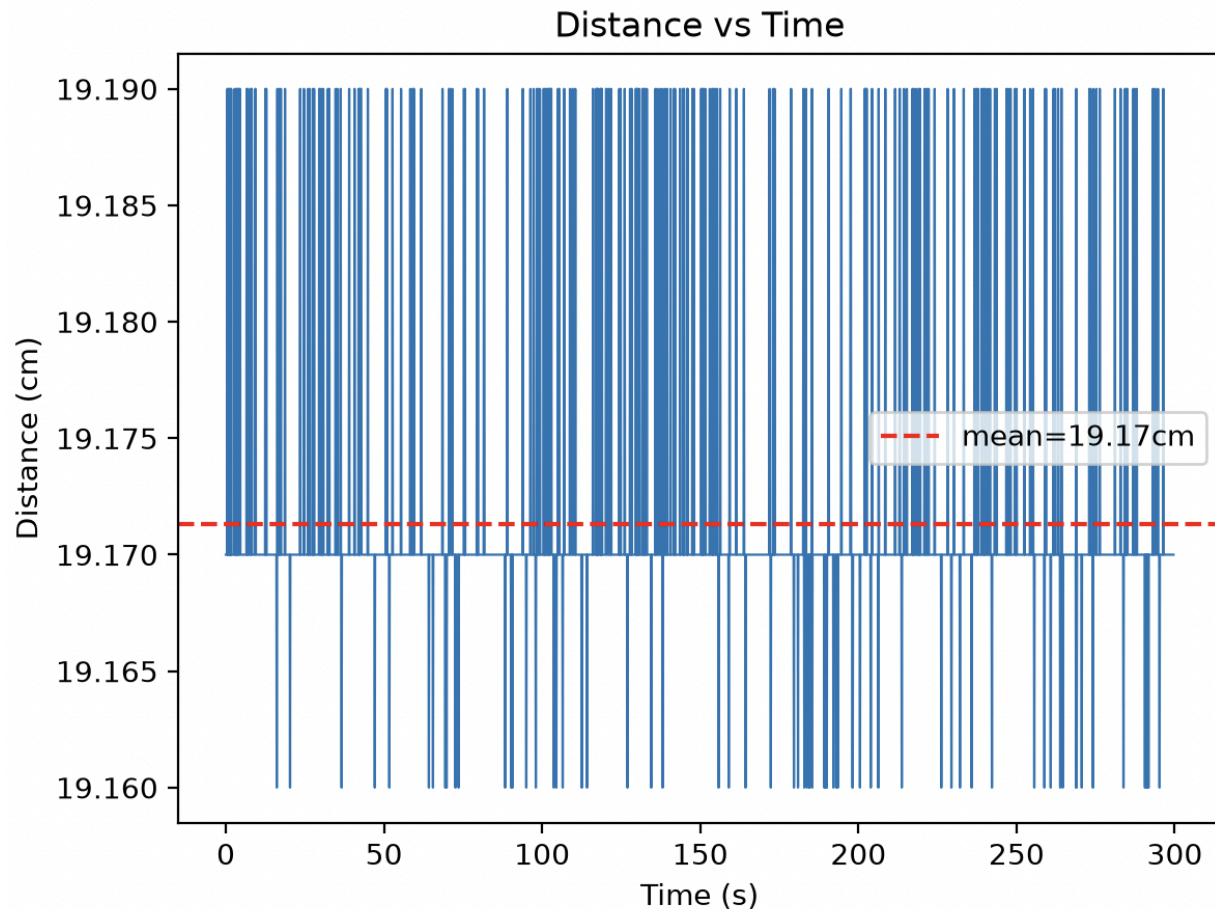
Line chart showing signed error (cm) on the y-axis against true distance (cm) on the x-axis, from the table above. Shows the sensor overshooting at short range (positive error, 5–10 cm), crossing zero around 12–13 cm, then increasingly undershooting at longer range (negative error, growing to -1.00 cm at 30 cm).

Noise Standard Deviation versus True Distance Chart

Noise Standard Deviation vs. True Distance (cm)



Line chart showing noise standard deviation (cm) on the y-axis against true distance (cm) on the x-axis. Shows noise staying low and flat from 5–15 cm (0.006–0.010 cm), then jumping sharply at 20 cm and staying elevated through 30 cm (0.29–0.34 cm).



Distance-vs-time plot over the full 5-minute stability test, showing a flat, tightly-clustered line with no visible upward or downward trend, which is consistent with the near zero drift rate and very low noise reported above.

Findings

The sensor's error was not constant in direction or magnitude across the tested range. At short distances (5–10 cm), the sensor consistently reads slightly high (+0.17 to +0.37 cm). Starting around 15 cm, this flipped — the sensor consistently read low, and that negative error grew larger with distance, reaching –1.00 cm at 30 cm. This suggests a distance dependent bias rather than a single fixed offset, with error magnitude roughly tracking distance beyond the 10–15 cm range.

Noise behaved differently than error. At close range (5–15 cm), noise was extremely low (0.006–0.010 cm std dev). The sensor was highly repeatable at short distances. Starting at 20 cm, noise jumped by roughly 30–50x (0.29–0.34 cm std) and stayed elevated through 30 cm. This points to a distinct transition in measurement reliability somewhere between 15 and 20 cm, rather than a smooth increase in noise with distance.

Notably, the stability test, ran at 20 cm, showed very low noise (0.006 cm std) and basically no drift (-0.0002 cm/min) over 5 minutes. This is a puzzling contrast: both tests used a target resting on the table, yet the accuracy sweep reading at 20 cm showed roughly 50x more noise than the stability test at effectively the same distance. Since target positioning wasn't the variable, other explanations are more likely: the target may not have been perfectly re-aligned (angle/position) each time it was repositioned between the 5, 10, 15, 20, 25, and 30 cm measurements during the sweep, whereas the stability test involved setting the target once and never touching it again; alternatively, the short 10-second sampling window in the accuracy sweep may not have been long enough to average out occasional multipath/reflection effects that a longer 5-minute window would smooth out. This discrepancy is itself a useful finding since it suggests the sensor's true intrinsic noise floor is closer to what the stability test showed (0.006 cm). Also the higher noise in the accuracy sweep at 20 cm+ may be a result of the test protocol rather than the sensor itself.

Next Steps

With more time, the following would help clarify the findings above:

- **Isolate alignment from sensor noise:** repeat the 20–30 cm accuracy points using a fixed jig or marked position so the target's angle and placement are identical every time it's repositioned, to check whether the elevated noise at these distances persists or was an artifact of small alignment differences between test runs.
- **Run longer accuracy samples:** extend each accuracy sweep measurement from 10 seconds to something closer to a minute, to see whether the noise at 20 cm+ drops toward the 0.006 cm level seen in the stability test once more samples are averaged.
- **Investigate the 15–20 cm transition:** take more precise measurements between 15 and 20 cm to pinpoint where the sign flip in error and the jump in noise actually occur.
- **Angle sensitivity:** test how readings change as the target is tilted relative to the sensor face, since ultrasonic sensors are known to be sensitive to angle of incidence.
- **Longer stability runs at multiple distances:** repeat the 5-minute stability test at 10 cm and 30 cm to see whether drift behavior changes with distance, not just at 20 cm.
- **Compare against a second sensor unit:** run the same protocol on a second HC-SR04 to see whether this offset/noise pattern is characteristic of the sensor model or specific to this individual unit.